

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

*I Moshitha Kolanjiyappan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

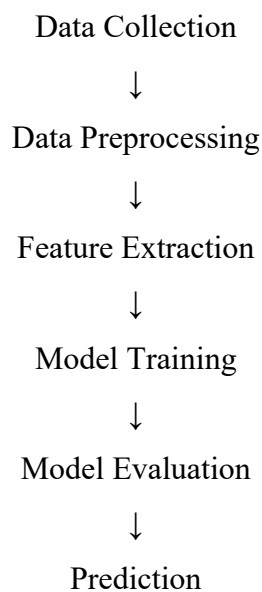
Signature of the student: Moshitha K

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process.

(5 Marks)



- Data Collection: Collect data from sensors, databases, or websites.
- Data Preprocessing: Clean data, remove missing values, and normalize data.
- Feature Extraction: Select important features that help in prediction.
- Model Training: Train the machine learning model using training data.
- Model Evaluation: Test the model using accuracy, precision, and recall.

Supervised Learning

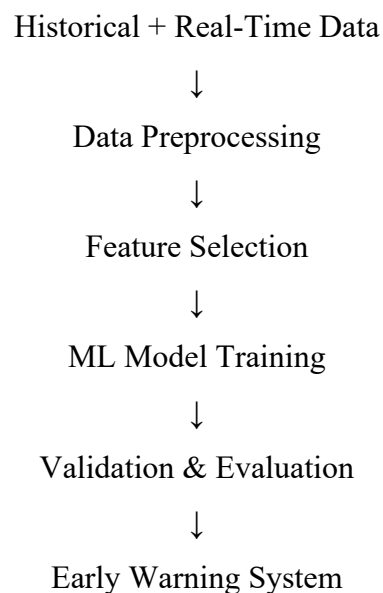
- Uses labelled data (input + correct output).
- Example: Email Spam Detection.

Unsupervised Learning

- Uses unlabelled data.
- Example: Customer Segmentation using K-Means clustering.

Difference: Supervised learning predicts known outputs, while unsupervised learning discovers hidden patterns.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)



- Data Sources: Weather data, rainfall, river levels, seismic sensor data, satellite images.
- Model Selection: Neural Networks (LSTM) are suitable because they can learn patterns from time-series data like rainfall and earthquakes.
- Preprocessing: Remove missing values, normalize data, and eliminate noise.

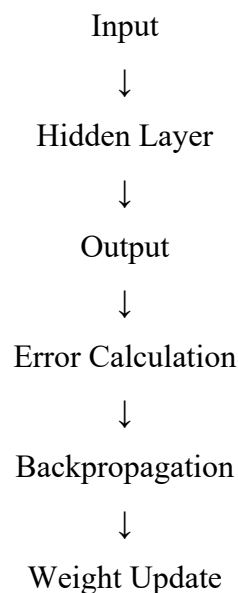
- Feature Selection: Select rainfall, temperature, river level, soil moisture, and seismic activity.
- Training & Validation: Train the model using historical data and validate using unseen data.
- Evaluation: Measure accuracy, precision, recall, and F1-score.

Applications

- Predict floods and earthquakes.
- Provide early warnings.
- Help authorities in evacuation planning.
- Support risk assessment and emergency response.

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model.

(5 Marks)



Backpropagation is a learning algorithm used to update the weights of a neural network.

Steps

1. Forward Pass: Inputs pass through the network to produce an output.
2. Error Calculation: Compare predicted output with the actual output.
3. Backward Pass: Propagate the error backward through the network.
4. Weight Update: Adjust weights using gradient descent to reduce the error.

Example

- Actual output = 1
- Predicted output = 0.8
- Error = $1 - 0.8 = 0.2$
- The weights are updated to reduce this error in the next iteration.

Role of Learning Rate

- High learning rate: Faster learning but may skip the optimal solution.
- Low learning rate: Slower learning but gives better accuracy.

Role of Activation Functions

- Introduce non-linearity into the network.
- Common functions: Sigmoid, ReLU, and Tanh.
- They improve convergence and enable the network to learn complex patterns.

Backpropagation continuously updates weights to minimize prediction error and improve the accuracy of the neural network.

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution.

(5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

Select individuals with the highest fitness values.

Selected Population:

- Example 5 (Fitness = 80)
- Example 3 (Fitness = 75)
- Example 4 (Fitness = 65)
- Combine the features of the selected parents to create new offspring.

Parents

Offspring

Example 5 + Example 3 High Speed, High Accuracy

Example 3 + Example 4 Medium Speed, High Accuracy

- These offspring inherit the best characteristics of their parents.

Randomly modify one feature to improve diversity.

Example:

- Medium Speed, High Accuracy → High Speed, High Accuracy

Mutation helps the algorithm explore new possible solutions and avoid getting stuck in local optima.

Solution	Speed	Accuracy	Estimated Fitness
Offspring 1	High	High	90
Offspring 2	Medium	High	82

Speed	Accuracy	Fitness	Suitable
High	High	90	Yes

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

Example	Actual Output	Predicted Output
1	Yes	Yes
2	Yes	No
3	No	No
4	Yes	Yes

Example Actual Output Predicted Output

5 No Yes

Neural Network Training using Backpropagation

Input Layer

(Age, BP, Glucose, Lifestyle)



Hidden Layer



Output Layer

(Disease: Yes / No)



Error Calculation



Backpropagation



Weight Update

- ☐ Forward Pass: Input features are passed through the network to predict the output.
- ☐ Error Calculation: Compare predicted output with the actual output.
- ☐ Backpropagation: Error is propagated backward to adjust weights.
- ☐ Weight Update: Weights are updated repeatedly until the error is minimized.

From the dataset:

- True Positive (TP): Actual = Yes, Predicted = Yes = 2
- True Negative (TN): Actual = No, Predicted = No = 1
- False Positive (FP): Actual = No, Predicted = Yes = 1
- False Negative (FN): Actual = Yes, Predicted = No = 1

Preprocessing

- Remove missing or incorrect values.
- Normalize Age, Blood Pressure, and Glucose values.
- Convert categorical values (High, Medium, Low) into numerical values.

Feature Selection

Select important features such as:

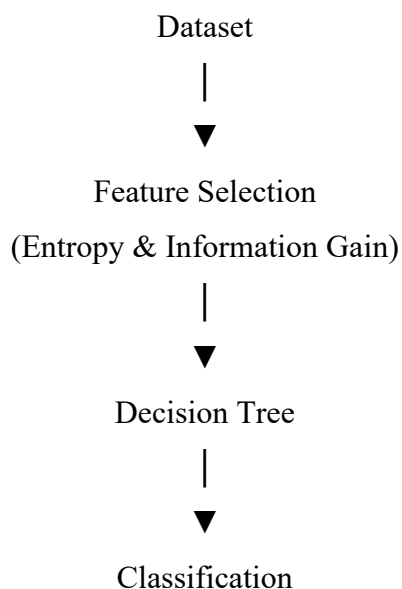
- Age
- Blood Pressure
- Glucose Level
- Lifestyle Risk

This reduces unnecessary data and improves model performance.

6. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models.

(5 Marks)

Diagram



- ☐ A Decision Tree is a supervised learning algorithm used for classification and prediction.
- ☐ Entropy measures the impurity of data.
- ☐ Information Gain selects the feature with the highest gain as the root node.
- ☐ The tree is built by splitting the dataset based on the best feature.
- ☐ Pruning removes unnecessary branches to reduce overfitting.

Weather Play

Sunny No

Rainy Yes

Sunny No

Overcast Yes

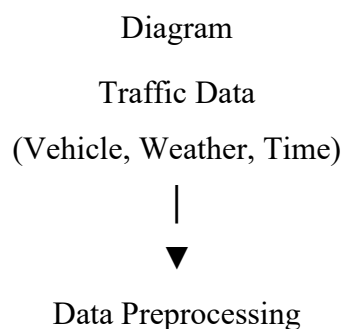
The feature Weather has the highest information gain, so it becomes the root node.

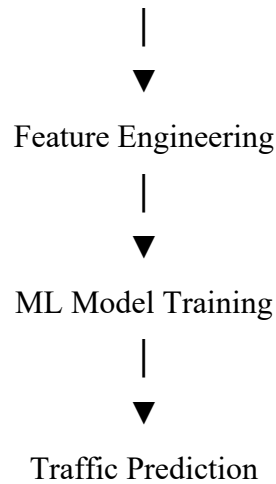
Overfitting Control

- Tree pruning
- Limiting tree depth
- Setting minimum samples per leaf

7. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making.

(5 Marks)

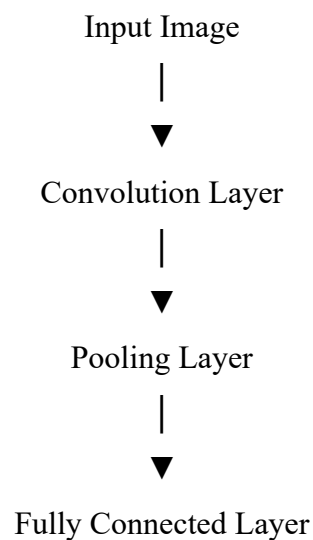


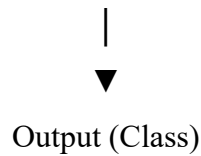


- ☐ Input Data: Vehicle density, time, weather, road type.
- ☐ Algorithm: Random Forest (handles multiple features and gives high accuracy).
- ☐ Preprocessing: Remove missing values, normalize data, encode categorical values.
- ☐ Feature Engineering: Select useful features like vehicle count, weather, and peak hours.
- ☐ Training: Train using historical traffic data.
- ☐ Validation: Test using unseen data.
- ☐ Deployment: Predict traffic congestion in real time for smart traffic signals and route planning.

8. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance.

(5 Marks)





- Convolution Layer: Extracts important image features using filters.
- Pooling Layer: Reduces image size and computation.
- Fully Connected Layer: Classifies the extracted features into output classes.

Feature Maps

- Filters detect edges, textures, and shapes to generate feature maps.

Backpropagation

- Computes prediction error.
- Updates filter weights using gradient descent.
- Repeats until error is minimized.

Example

Input image → CNN → Output: Cat or Dog

Factors Affecting Performance

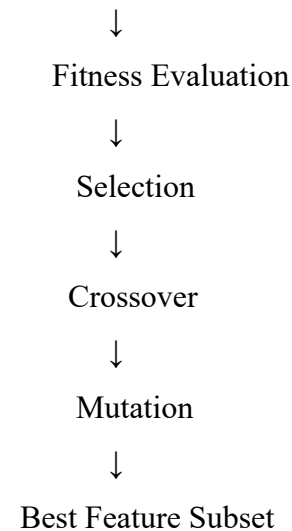
- Dataset quality
- Number of training images
- Learning rate
- Number of epochs
- CNN architecture (layers and filters)

9. Consider the application of **Genetic Algorithms for feature selection** in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better

feature subsets, and discuss how this improves model accuracy and reduces complexity.

(5 Marks)

Feature Subsets (Chromosomes)



- Chromosome: Each chromosome represents a subset of features using binary values (1 = selected, 0 = not selected).
Example: 10110 (Features 1, 3, and 4 selected).
- Fitness Evaluation: Measures model accuracy using the selected features.
- Selection: Chooses chromosomes with the highest fitness.
- Crossover: Combines parent chromosomes to create better offspring.
- Mutation: Randomly changes a gene (0 ↔ 1) to introduce diversity.

Advantages

- Improves prediction accuracy.
- Removes unnecessary features.
- Reduces computation time and model complexity.

10. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their

importance in applications such as medical diagnosis and fraud detection.

(5 Marks)

	Predicted Yes	Predicted No
Actual Yes	TP = 40	FN = 10
Actual No	FP = 5	TN = 45

Metrics

- $\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total} = (40 + 45) / 100 = 85\%$
- $\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 40 / 45 = 88.9\%$
- $\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 40 / 50 = 80\%$
- $\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 84.2\%$
- ROC Curve: Graph showing the trade-off between True Positive Rate and False Positive Rate.

Applications

- Medical Diagnosis: High recall reduces missed disease cases.
- Fraud Detection: High precision reduces false fraud alerts.

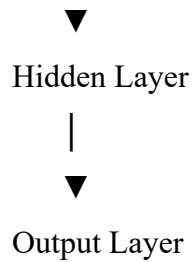
11. With the help of a neat diagram, explain the architecture of an Artificial Neural Network (ANN). Describe the roles of input, hidden, and output layers, and explain how weights and biases are used in computation. Illustrate how forward propagation works with a simple example, and discuss how overfitting can be controlled using techniques such as dropout and regularization.

(5 Marks)

Diagram

Input Layer

|



- Input Layer: Receives input features.
- Hidden Layer: Processes data using weights, biases, and activation functions.
- Output Layer: Produces the final prediction.

Weights and Biases

- Weights: Determine the importance of each input.
- Bias: Helps adjust the output for better learning.

Forward Propagation

1. Inputs are multiplied by weights.
2. Bias is added.
3. Activation function computes the output.
4. Output is passed to the next layer.

Example: Input (Age, BP) → Hidden Layer → Output (Disease: Yes/No).

Controlling Overfitting

- Dropout: Randomly disables neurons during training.
- Regularization: Reduces large weights to improve generalization.

12. Design a machine learning system for weather forecasting to predict rainfall using features such as temperature, humidity, wind speed, and atmospheric pressure. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time predictions.

(5 Marks)

Weather Data
(Temperature, Humidity,
Wind Speed, Pressure)



Data Preprocessing



Feature Engineering



ML Model Training



Validation & Testing



Rainfall Prediction

- ☐ Algorithm: Random Forest (high accuracy and handles multiple features well).
- ☐ Preprocessing: Remove missing values, normalize data, and remove noise.
- ☐ Feature Engineering: Select important features such as temperature, humidity, wind speed, and pressure.
- ☐ Training: Train the model using historical weather data.
- ☐ Validation: Test the model using unseen data.
- ☐ Deployment: Predict rainfall in real time for weather forecasting and disaster management.

13. In the context of Recurrent Neural Networks (RNNs), explain the architecture and how they handle sequential data. Describe the concept of hidden states and how information is passed through time steps. Illustrate with an example such as text prediction or time-series forecasting, and discuss challenges like vanishing gradients and possible solutions.

(5 Marks)

Input → Hidden State → Output



Input → Hidden State → Output



Input → Hidden State → Output

- RNN is used for sequential data such as text and time-series.
- Hidden State: Stores information from previous time steps and passes it to the next step.
- This helps the network remember past information.

Example

- Text Prediction: "I love" → predicts "AI".
- Time-Series Forecasting: Predicts future rainfall based on previous weather data.

Challenges

- Vanishing Gradient: Earlier information is forgotten during training.
- Solutions: Use LSTM or GRU networks, which remember information for longer periods.

14. Consider the use of Genetic Algorithms for optimization problems such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently.

Initial Population



Fitness Evaluation



Selection



Crossover



Mutation



Best Schedule

- Chromosome: Represents a task schedule.
- Fitness Function: Measures how efficiently tasks are completed (minimum time and cost).
- Selection: Chooses the best schedules.
- Crossover: Combines parent schedules to create better offspring.
- Mutation: Randomly changes a task assignment to improve diversity.

Advantages

- Finds near-optimal schedules.
- Reduces execution time.
- Improves resource utilization.

15. In the context of classification models, explain how performance evaluation is carried out using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Using a sample confusion matrix, demonstrate how each metric is calculated and discuss their importance in real-world applications such as spam detection and disease diagnosis.

(5 Marks)

	Predicted Yes	Predicted No
Actual Yes	TP = 40	FN = 10
Actual No	FP = 5	TN = 45

Metrics

- Accuracy = $(TP + TN) / \text{Total} = 85\%$
- Precision = $TP / (TP + FP) = 88.9\%$
- Recall = $TP / (TP + FN) = 80\%$
- F1-Score = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 84.2\%$
- ROC-AUC: Measures the model's ability to distinguish between positive and negative classes. A higher AUC indicates better performance.

Applications

- Spam Detection: High precision reduces incorrect spam classification.
- Disease Diagnosis: High recall ensures more patients with the disease are correctly identified.

Conclusion: These evaluation metrics help compare classification models and select the most reliable one for real-world applications.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand